Parent Guide

@twinklparents

We're excited to share this activity with you. If you are interested in finding more engaging, fun and interesting activities for you and your children, then check out these links to different areas of the **Twinkl Parents** website.

games

crafts

puzzles

experiments

word searches

What is this resource and how do I use it?

Your child will interpret information and place data into Venn diagrams and Carroll diagrams. This resource explains what each of these diagrams are and how they work. Your child can then have a go at the guided questions, before having a go at the independent ones, to build their confidence.

What skills does this practise?

- Interpreting Venn Diagrams
- Interpreting Carroll Diagrams
- Sorting Data

Further Activity Ideas and Suggestions

Your child can cement key skills such as **counting forwards and backwards in steps of 6, 7 and 9** or find which numbers are **greater than, less than or equal to** others. Try this **Christmas maths mystery** to inject some excitement into developing key skills.

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Statistics:

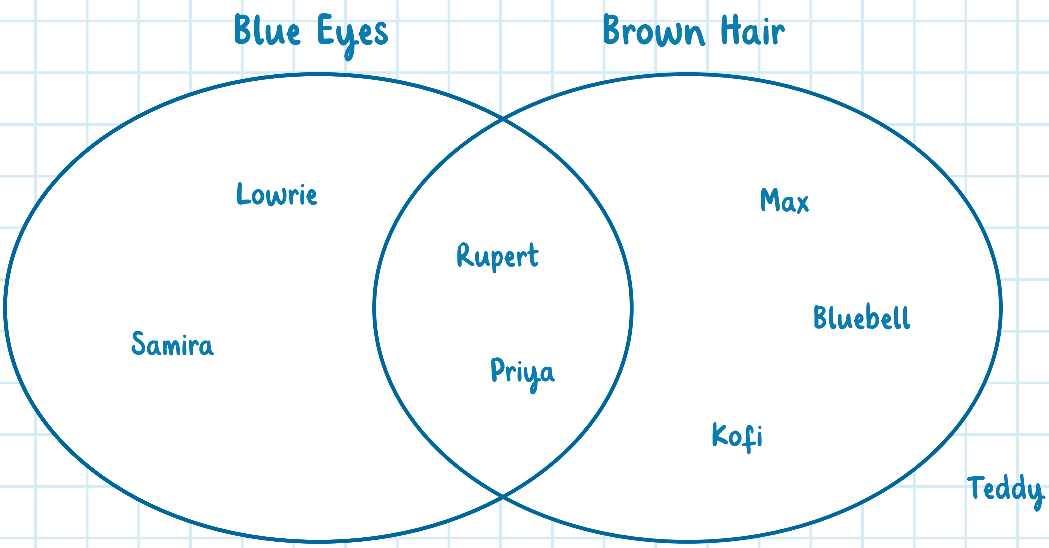
Venn Diagrams

What is a Venn diagram?

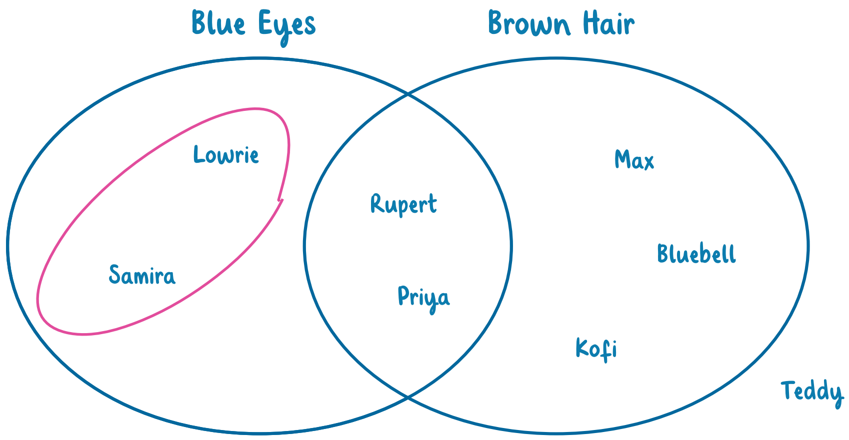
A Venn diagram is a diagram made up of overlapping circles. It's used to sort data according to its similarities and differences. Each circle or semicircle of the Venn diagram has a rule to sort information. The space where the two shapes overlap is used to place pieces of information that satisfy both rules. If a piece of information doesn't satisfy either of the rules, it is placed outside the shapes.

Have a look at this example.

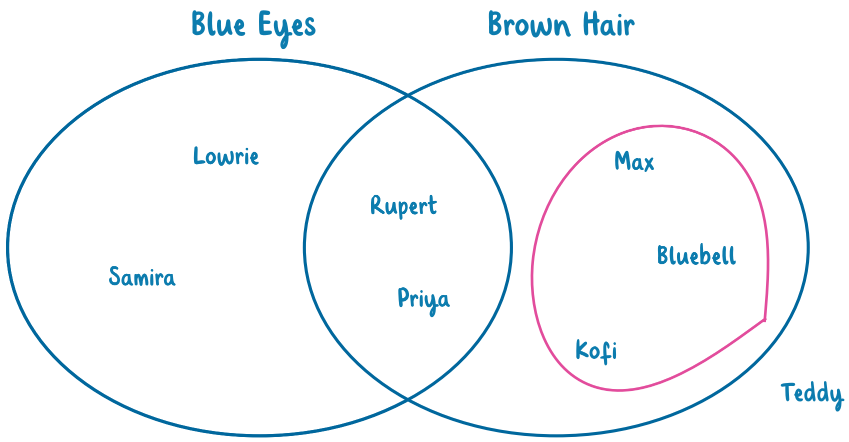
Hair Colour and Eye Colour



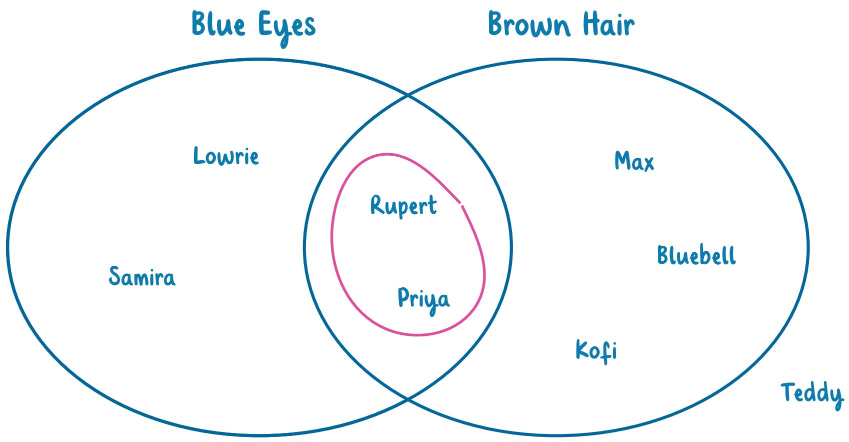
From the Venn diagram, we can see that there are two children who have blue eyes **but not** brown hair:



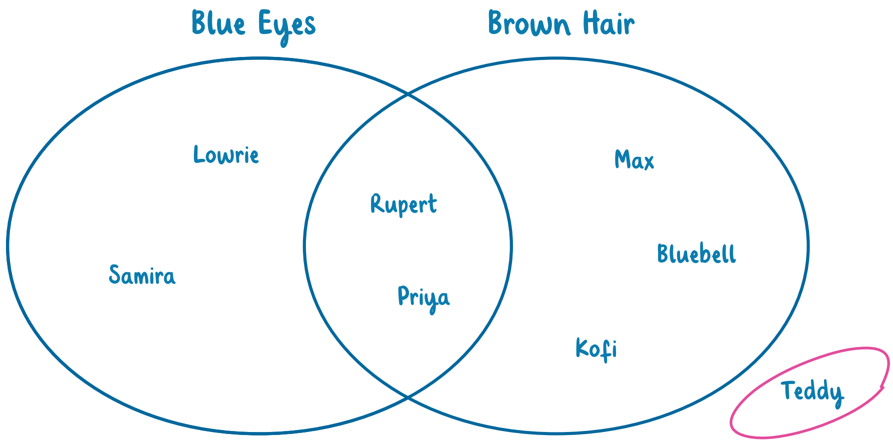
We can see that there are three children with brown hair **but not** blue eyes:



There are two children in the middle section, where the shapes overlap. This means that they have blue eyes **and** brown hair.



One child's name does not appear within the shapes. This means that they **do not** have blue eyes or brown hair.



Questions

Using the Venn diagram, we can answer these questions.

1. **How many children have blue eyes?**
   This means we have to look at the **whole** of the shape labelled 'blue eyes'.

   There are ☐ children with blue eyes.

2. **How many children do not have brown hair?**
   This means we have to count the number of children whose names **do not** appear in the shape labelled 'brown hair'.

   ☐ children do not have brown hair.

3. **Could this be Bluebell? Why or why not?**
   We can answer this by looking at where Bluebell's name appears in the Venn diagram.

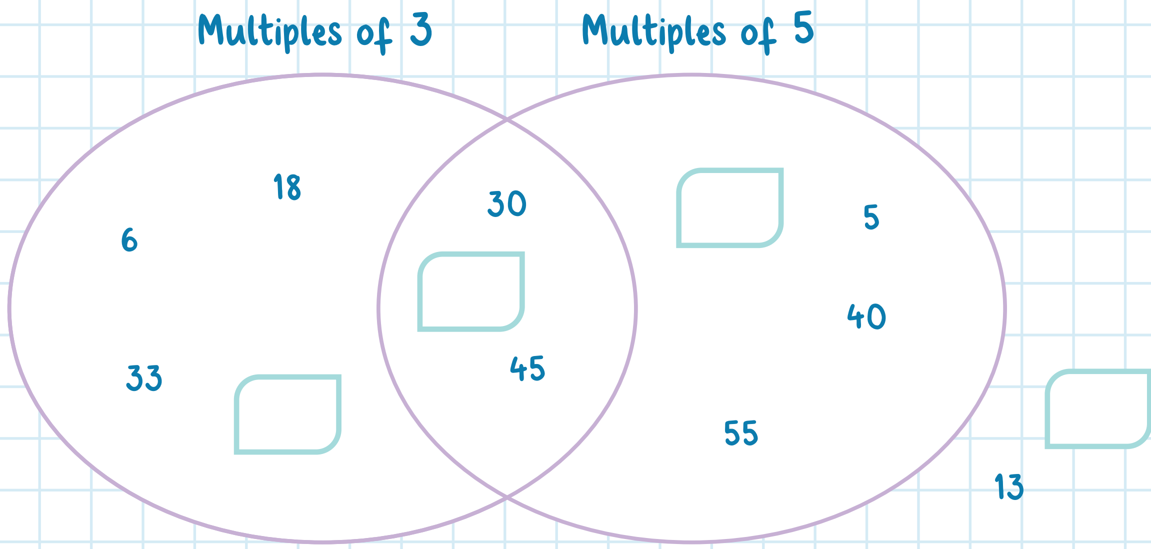
   ____

   ____

Multiples Venn Diagram

This Venn diagram shows numbers that are multiples of 3 and 5.

Multiples of 3 and 5



Questions

1. Put one number in each box to satisfy the rules.
2. How many numbers (including the ones you've added in) are multiples of 5 and 3?

   ____

3. Which numbers are not multiples of 3 or 5?

   ____

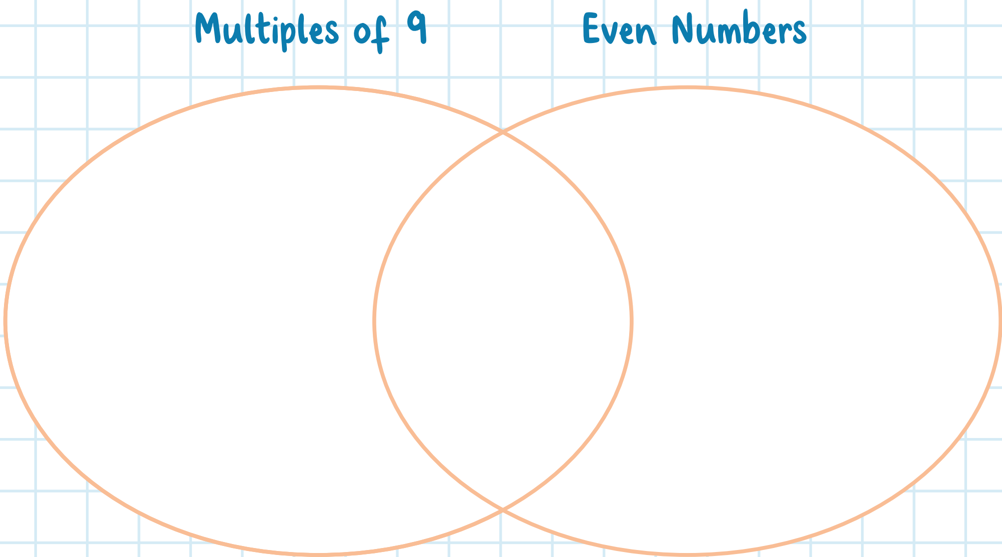
4. Olivia says, "There are more multiples of 5 than multiples of 3 in the Venn diagram."
   Is she correct?

   ____

5. Now, put these numbers into the Venn diagram below.

9	28	17	12	58	18
35	27	33	46	36	45

Multiples of 9 and Even Numbers



Statistics:

Carroll Diagrams

What is a Carroll diagram?

A Carroll diagram is set out in rectangles, like a data table. It's used to sort data according to traits and the categories require a definite 'yes' or 'no' answer. There are four different spaces to place data. To find out where to place data, we need to read the categories at the top, deciding which one our data will satisfy. Then, we need to read the categories down the side. The categories the data satisfies determine where it is placed within the Carroll diagram.

Have a look at this example.

	2 Legs	More than 2 Legs
Can Fly	eagle starling	bee ladybird
Can't Fly	penguin human	cat centipede

From the Carroll diagram, we can see that the animals that have **2 legs and can fly** are the eagle and the starling.

	2 Legs	More than 2 Legs
Can Fly	eagle starling	bee ladybird
Can't Fly	penguin human	cat centipede

We can see that the animals that have **2 legs and can't fly** are the penguin and the human.

	2 Legs	More than 2 Legs
Can Fly	eagle starling	bee ladybird
Can't Fly	penguin human	cat centipede

The animals with **more than 2 legs that can fly** are the bee and the ladybird.

	2 Legs	More than 2 Legs
Can Fly	eagle starling	bee ladybird
Can't Fly	penguin human	cat centipede

Finally, the animals with more than **2 legs that can't fly** are the cat and the centipede.

	2 Legs	More than 2 Legs
Can Fly	eagle starling	bee ladybird
Can't Fly	penguin human	cat centipede

Questions

Using the Carroll diagram, we can answer these questions.

1. **Which animals in the Carroll diagram have more than 2 legs?**
   To answer this, we have to look at all of the sections in the column 'More than 2 Legs'.

   The animals with more than 2 legs are ______, ______, ______ and ______.

2. **How many animals can't fly?**
   To find the answer, we have to look at all of the sections in the 'Can't Fly' row.

   There are ______ animals that can't fly.

3. **Natalie places 'lion' in the same box as 'centipede'. Has she put it in the right place?**
   To answer this, we need to look at the box that 'centipede' is in and find out what categories it satisfies. Then, we can decide if a lion satisfies those same categories.

   ____

   ____

Multiples Carroll Diagram

This Carroll diagram shows which numbers are multiples.

	Multiple of 6	Not a Multiple of 6
Multiple of 5	30 60	25 15 45
Not a Multiple of 5	36 12	21 49

Questions

1. Write down the numbers from the Carroll diagram that are multiples of 5.

2. Gupta says, " The numbers in the top right-hand box will always end with a 5."
Is she right? Explain your answer.

3. Shane says, "All of the numbers not in the 6 x table will be in the bottom right-hand box."
Explain why he is not right.

4. Calla places the number 120 in the top left-hand box. Has she put it in the right place?

5. Place the following numbers into the Carroll diagram below.

24	50	12	42	4	16
18	40	30	26	36	23

Multiples of 4 and 6

	Multiple of 6	Not a Multiple of 6
Multiple of 4		
Not a Multiple of 4		

Statistics: Venn Diagrams and Carroll Diagrams Answers

1. How many children have blue eyes?

   There are **4** children with blue eyes.

2. How many children do not have brown hair?

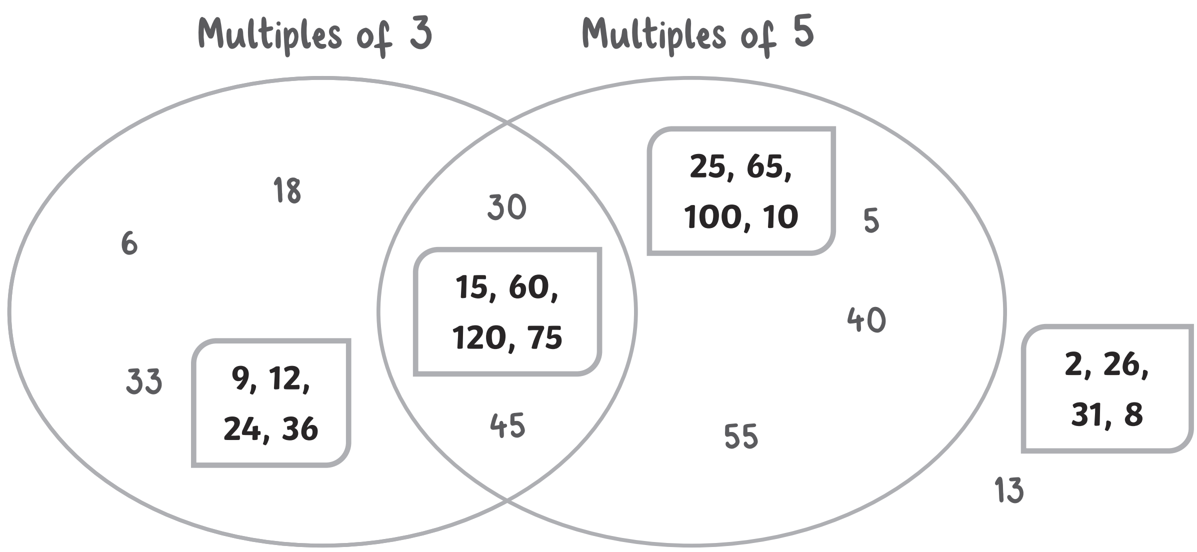
   3 children do not have brown hair.

3. Could this be Bluebell? Why or why not?
   This could be Bluebell. Bluebell has brown hair but not blue eyes. This girl has brown hair and brown eyes.

Multiples Venn Diagram

1. Put one number in each box to satisfy the rules.
   There are lots of answers; we have put some examples into the boxes.

Multiples of 3 and 5



2. How many numbers (including the ones you've added in) are multiples of 5 and 3?
11

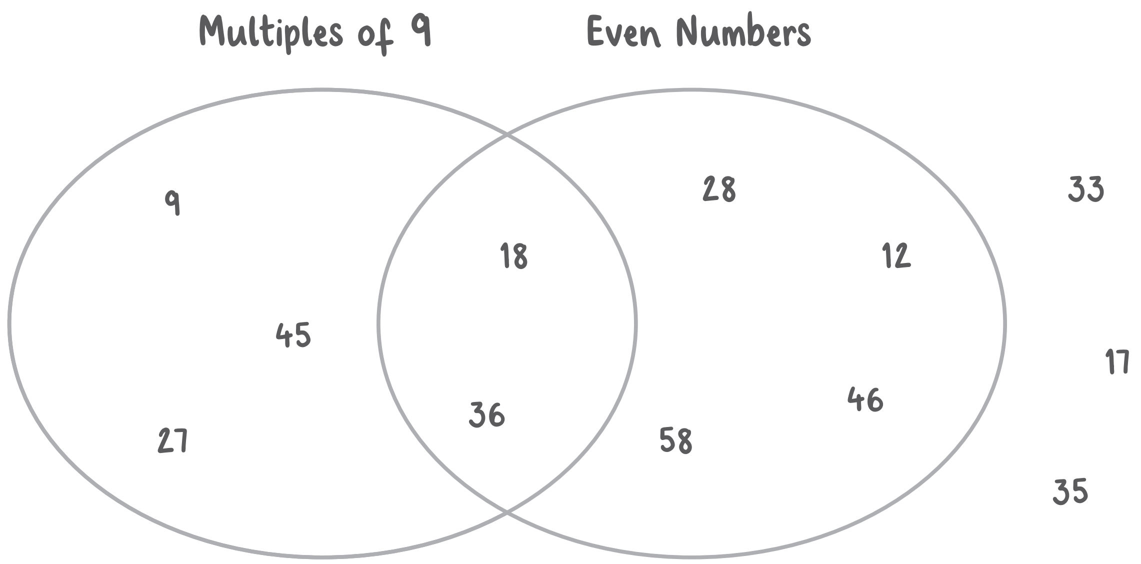
3. Which numbers are not multiples of 3 or 5?
13 (and the one you added in the box)

4. Olivia says, "There are more multiples of 5 than multiples of 3 in the Venn diagram."
Is she correct?
There are 7 multiples of 5 in the Venn diagram. There are also 7 multiples of 3. Olivia is incorrect.

5. Now, put these numbers into the Venn diagram below.

9	28	17	12	58	18
35	27	33	46	36	45

Multiples of 9 and Even Numbers



Carroll Diagrams

1. Which animals in the Carroll diagram have more than 2 legs?
   The animals with more than 2 legs are **bee**, **ladybird**, **cat** and **centipede**.
2. How many animals can't fly?
   There are **4** animals that can't fly.
3. Natalie places 'lion' in the same box as 'centipede'. Has she put it in the right place?
   Yes, because a lion has more than 2 legs and can't fly, the same as a centipede.

Multiples Carroll Diagram

1. Write down the numbers from the Carroll diagram that are multiples of 5.
   30, 60, 15, 25, 45
2. Gupta says, " The numbers in the top right-hand box will always end with a 5."
   Is she right?
   Explain your answer.
   Gupta is wrong. Numbers that are in the 5 x table but not in the 6 x table will also appear, for example, 10, 20, 40 and so on.
3. Shane says, "All of the numbers not in the 6 x table will be in the bottom right-hand box."
   Explain why he is not right.
   Shane is not right because the bottom right-hand box is for numbers that are not multiples of 6 or 5. However, some numbers will not be multiples of 6 but will be multiples of 5. These numbers will appear in the top right-hand box.
4. Calla places the number 120 in the top left-hand box. Has she put it in the right place?
   Yes she has because 120 is a multiple of 5 and a multiple of 6.

5. Place the following numbers into the Carroll diagram below.

24	50	12	42	4	16
18	40	30	26	36	23

Multiples of 4 and 6

	Multiple of 6	Not a Multiple of 6
Multiple of 4	24 36 12	16 40 4
Not a Multiple of 4	42 30 18	50 26 23